

Increased platelet to red blood cell transfusion ratio associated with acute kidney injury in children with life-threatening bleeding

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Abstract

Introduction: Transfusion may increase the risk of organ failure through immunomodulatory effects. The primary objective of this study was to assess for patient or transfusion-related factors that are independently associated with the risk of acute kidney injury (AKI) and acute respiratory distress syndrome (ARDS) in a cohort of children with life-threatening bleeding from all etiologies.

Methods: In a secondary analysis of the prospective observational massive transfusion in children (MATIC) study, multivariable logistic regression was performed in an adjusted analysis to determine if blood product ratios or deficits were independently associated with AKI or ARDS in children with life-threatening bleeding.

Results: There were 449 children included with a median (interquartile range, IQR) age of 7.3 years (1.7–14.7). Within 5 days of the life-threatening bleeding event, AKI occurred in 18.5% and ARDS occurred in 20.3% of the subjects. Every 10% increase in the platelet to red blood cell transfusion ratio is independently associated with a 12.7% increase in the odds of AKI (adjusted odds ratio

Abbreviations: AKI, acute kidney injury; ARDS, acute respiratory distress syndrome; BRMs, biological response modifiers; ECMO, extracorporeal membrane oxygenation; INR, International Normalized Ratio; IPW, inverse probability weighted; IQR, interquartile range; LTB, life-threatening bleeding; MATIC, massive transfusion in children study; PATCH, platelet Transfusion in Cerebral Hemorrhage Trial; pRBCs, packed red blood cells; PRISM, pediatric risk of mortality score; PROPPR, Pragmatic randomized optimal platelet and plasma ratios trial; PT, prothrombin time; PTT, partial thromboplastin time; RBCs, red blood cells.

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1.127; 95% confidence interval 1.025–1.239; *p*-value .013). Subjects with operative or medical etiologies were independently associated with an increased risk of AKI compared to those with traumatic injury. No transfusion-related variables were independently associated with the risk of developing ARDS.

Conclusion: The use of increased platelet to red blood cell transfusion ratios in children with life-threatening bleeding of any etiology may increase the risk of AKI but not ARDS. Prospective trials are needed to determine if increased platelet use in this cohort increases the risk of AKI to examine possible mechanisms.

KEYWORDS

acute kidney injury, acute respiratory distress syndrome, balanced transfusion, pediatric patients, plasma deficit, plasma:Red blood cell, platelet deficit, platelet:Red blood cell

1 | INTRODUCTION

While life-threatening bleeding is relatively uncommon in children, the risk of mortality in this setting is high,¹ and optimal resuscitation practice has not yet been determined. Death from traumatic hemorrhage is the most common cause of preventable death after injury. Children can also have life-threatening bleeding from operative and medical causes that are also associated with high mortality.¹ Hemostatic resuscitation, defined as the use of blood products in a balanced approach to simultaneously and rapidly address shock, hemostatic dysfunction, and endothelial injury in patients with life-threatening bleeding,² aims at reducing death from hemorrhage. This is accomplished with either whole blood or blood components in a balanced transfusion strategy. Balanced strategies can be determined according to the ratio provided (volume plasma:volume RBC or volume platelet:volume RBC) or as a deficit (volume RBC-volume plasma or volume RBC-volume platelet).^{3,4} The current standard of care for pediatric hemostatic resuscitation is based on adult trauma literature^{5–7}; the data supporting hemostatic resuscitation in children is less robust.^{8,9}

Many clinical trials and medical advances begin in adult medicine, are applied to pediatric patients, and then are studied and adapted to better suit the pediatric population; this is no different from life-threatening bleeding research in children. However, pediatric patients have many developmental differences from adult patients including their pathophysiologic response to severe bleeding, age-related maturation of the hemostatic system, and increased risk of head injury thus their response to transfusion strategies may vary compared to adults^{1,10,11}; it is necessary to develop pediatric-specific data. Further, recent data in children suggests that mortality and effectiveness of hemostatic resuscitation

strategies depend on the etiology of bleeding,^{4,12} though few studies include children with various causes of bleeding (operative, medical, traumatic).

Adverse effects of transfusion may occur due to transfusion-related immunomodulation,¹³ which can cause a hyperinflammatory and pro-coagulant state that can increase the risk of organ failure.¹³ The long-term outcomes for children that develop these conditions can increase the risk of chronic renal and pulmonary failure^{14,15} and therefore it is important to better understand the impact of transfusion on these outcomes in children. The primary objective of this study was to assess for patient or transfusion-related factors that are independently associated with the risk of acute kidney injury (AKI) and acute respiratory distress syndrome (ARDS) in a cohort of children with life-threatening bleeding.

2 | METHODS

A secondary analysis was performed from data collected in the massive transfusion in children (MATIC) study. In brief, MATIC was a prospective observational study of children with life-threatening bleeding from all etiologies conducted from 2014 to 2018 at 24 centers internationally.¹ Life-threatening bleeding was defined as a subject who received 40 mL/kg or more of all blood products or activation of a massive transfusion protocol with at least initiation of transfusion of blood products. The second definition is to include children who may have died before 40 mL/kg of blood in total could be given. Subjects in the study were categorized with either trauma, operative, or medical etiologies of life-threatening bleeding. The blood product ratios and deficits used for analysis were restricted to those transfused during the life-threatening bleeding event. Products used before or after

TABLE 1 Demographics of children with life-threatening bleeding.

Measures	Overall population <i>N</i> = 449
Sex	
Male	247 (55%)
Age (years)	7.3 [1.7, 14.7]
Race	
White	242 (63.7%)
Black	109 (28.7%)
Other	29 (7.6%)
Hispanic/Latino	43 (11%)
Blood group	
A	157 (36%)
B	65 (14.9%)
AB	12 (2.8%)
O	202 (46.3%)
Rhesus factor	
+	359 (82.7%)
LTB location	
Cardiovascular intensive care unit	39 (8.7%)
Emergency department	160 (35.6%)
Inpatient floor	4 (0.9%)
Neonatal intensive care unit	10 (2.2%)
Operating room	125 (27.8%)
Pediatric intensive care unit	105 (23.4%)
Other location	6 (1.3%)
Pediatric risk of mortality III (PRISM score)	12.0 [6.0, 22.0]
Resuscitation	
Mechanical ventilation	255 (59%)
ECMO	36 (8%)
Injury severity score	29.0 [20.5, 38.0]
Vital signs and laboratory values prior to LTB	
Age-adjusted hypotension	255 (57.6%)
Tachycardia	263 (62.3%)
Lowest body temperature (°C)	36.2 [35.2, 36.6]
Highest body temperature (°C)	36.9 [36.3, 37.5]
Lowest pH	7.2 [7.1, 7.3]
Highest serum creatinine (mg/dL)	0.6 [0.4, 1.0]
Lowest hemoglobin (g/dL)	9.6 [7.8, 11.7]
Lowest white blood cell count (K/mm ³)	10.6 [6.1, 17.6]
Lowest platelet count (×10 ⁹ /L)	152.0 [76.0, 245.0]

(Continues)

TABLE 1 (Continued)

Measures	Overall population <i>N</i> = 449
Highest INR	1.5 [1.2, 2.0]
Highest PT (s)	17.7 [14.9, 22.5]
Highest PTT (s)	43.0 [31.6, 87.5]
Fibrinogen (mg/dL)	190.0 [135.0, 259.5]
AKI	83/449 (18.5%)
ARDS	91/449 (20.3%)
28-day mortality	168/445 (37.8%)
Transfused blood products	
Total Volume Transfused (mL/kg)	61.3 [29.0, 122.5]
pRBCs (mL/kg)	30.2 [15.5, 62.3]
Plasma (mL/kg)	17.1 [4.8, 35.6]
Platelets (mL/kg)	7.3 [0.0, 19.8]
Received cryoprecipitate (%)	152 (33.9%)
Plasma:pRBC transfusion ratio	0.5 [0.2, 0.9]
Platelet:pRBC transfusion ratio	0.2 [0.0, 0.4]

Note: Data presented as either *N* (%) or median (interquartile range).

Abbreviations: AKI, acute kidney injury; ARDS, acute respiratory distress syndrome; ECMO, extracorporeal membrane oxygenation; INR, international normalized ratio; LTB, life-threatening bleeding; pRBCs, packed red blood cells; PT, prothrombin time; PTT, partial thromboplastin time; Total Volume Transfused, platelets + pRBC + plasma + cryoprecipitate.

the life-threatening bleeding event were not used to calculate ratios or deficits. Additional details regarding the methods and data collected are summarized in the primary manuscript.¹

In this study, the primary outcomes were AKI and ARDS. AKI was defined according to KDIGO (Kidney Disease: Improving Global Outcomes) criteria.¹⁶ The definition of ARDS was according to the Berlin criteria¹⁷; the pediatric ARDS definition was not used since it was developed after data collection started for the MATIC study.

The characteristics of the cohort were summarized as median with interquartile range (IQR) for continuous variables and percentages for categorical variables. Kruskal-Wallis and chi-square tests were used for unadjusted comparison of continuous and categorical variables, respectively. The trial observation period for the occurrence of AKI and ARDS was 5 days from admission; however, these were captured as binary occurrences within that time-window. Thus, mortality is a competing risk in that time-window, but the timing of AKI and ARDS is unknown precluding the use of time-to-event competing risk modeling. To measure the association of plasma and platelet ratios and deficits with AKI and

ARDS while accounting for the potential of early death within the five-day window, we utilized inverse probability weighted (IPW) multivariable logistic model with robust standard error implemented in the survey-weighted generalized linear model function built in R package survey (Supplemental Methods and Figure S1). We used an ordinal logistic regression¹⁸ model to compute the IPW for each patient based on the day of death up to 5 or beyond to allow down weighting of patients with earlier deaths and less observed time to develop AKI or ARDS (Supplemental Methods and Table S3). These weights were applied to each patient in the multivariable logistic regression models with AKI or ARDS as the outcome of interest. The AKI model was adjusted for mechanical ventilation, extracorporeal membrane oxygenation (ECMO), clinical group, sex, age, whether cardiac arrest occurred, platelet count, pediatric risk of mortality (PRISM) score, and total amount of blood products transfused (including pRBCs, platelets, plasma, and cryoprecipitate). Hemoglobin was removed from the model due to collinearity issue ($VIF = 13$). The ARDS model was adjusted for the same variables as the AKI model except whether the patient had cardiac arrest, platelet count, and total amount of blood products transfused. Feature selection was based on univariate logistic regression, backward stepwise feature selection with AIC as the criteria, and clinical expertise. Variables significant in the univariable model were included in the multivariable model, except those missing greater than 35%. Odds ratios were interpreted for every 0.1 (or 10%) unit change instead of the default 1 unit for ratio variables as they are bounded between 0 and 1. Collinearity was checked for all adjusted covariates. Hosmer and Lemeshow goodness of fit test was also conducted for all models. All analyses were conducted using R (version 4.3.0, Vienna, Austria).

3 | RESULTS

A total of 449 children with life-threatening bleeding met inclusion criteria. Of the patients analyzed, median (IQR) age was 7.3 years (1.7–14.7) and 55% were male (Table 1). For those with documented race and ethnicity, 63.7% were White, 28.7% were Black, and 11% were Hispanic. Within 5 days of the life-threatening bleeding event, AKI occurred in 18.5% and ARDS occurred in 20.3% of the subjects. Prior to the life-threatening bleeding, subjects who developed AKI had lower hemoglobin, serum creatinine, platelet count, and higher rates of ECMO compared to those who did not develop AKI (Table 2). Prior to life-threatening bleeding, subjects who developed ARDS had increased heart rate, serum creatinine, PRISM scores, and rates of mechanical ventilation and ECMO compared

to those without ARDS (Table 3). In those with traumatic injury, the injury severity score was higher in subjects with ARDS compared to those without ARDS.

Univariate associations with AKI and ARDS are reported in the Tables S1 and S2. Both AKI and ARDS multivariable logistic regression models passed the Hosmer and Lemeshow goodness of fit test with a p -value of .111 and .656 respectively. The results of the mortality-weighted multivariable logistic regression model (Table 4) indicate that for every 10% increase in the ratio of transfused platelets to packed red blood cells, the estimated odds of AKI increased by 12.7% (adjusted odds ratio = 1.127; 95% CI 1.025–1.239; p -value .013). Operative or medical etiology of bleeding was independently associated with an increased risk of AKI compared to traumatic injury. The transfused plasma to red blood cell ratio was not associated with an increased odds of developing AKI. Additionally, no transfusion ratios, including plasma to RBC ratio, platelet to RBC ratio, plasma deficit, or platelet deficit, were independently associated with increased odds of developing ARDS (Table 5).

4 | DISCUSSION

In this secondary analysis of the MATIC study, there was an increased risk of AKI for children who receive transfusions at increased ratios of platelets to RBCs during their life-threatening bleeding event. Additional studies examining potential adverse effects of hemostatic resuscitation in children with life-threatening bleeding are needed to inform pediatric-specific guidelines.

There are existing studies that have investigated the impact of platelet transfusion on morbidity in both adults and children. A multicenter randomized control trial examined platelet transfusion for prophylaxis or treatment of bleeding in premature infants at risk for intraventricular hemorrhage. Increased use of platelets at higher versus lower transfusion thresholds revealed an increased risk of mortality or severe bleeding compared to the more restrictive use of platelets in this population.¹⁹ These results were similar to the platelet transfusion in cerebral hemorrhage (PATCH) trial in adults with hemorrhagic stroke while on dual antiplatelet agents, where the use of platelets compared to placebo caused increased morbidity and mortality at 90 days.²⁰ The pragmatic randomized optimal platelet and plasma ratios (PROPPR) trial in adult trauma patients, that examined platelet and plasma to RBC transfusion ratios, did not report an increased risk of AKI or ARDS or any other organ failure outcome or adverse effect.⁷ No other trials of hemostatic resuscitation with either components or whole blood have been conducted outside of trauma populations.

TABLE 2 Demographics of children with life-threatening bleeding with and without acute kidney injury (AKI).

Measures	No AKI (N = 366)	AKI (N = 83)	p-value
Sex	-	-	-
Male	202 (55.2%)	45 (54.2%)	.969
Age (years)	7.7 [2.0, 14.7]	4.4 [0.4, 14.5]	.085
Race	-	-	.295
White	195 (64.1%)	47 (61.8%)	-
Black	89 (29.3%)	20 (26.3%)	-
Other	20 (6.6%)	9 (11.8%)	-
Hispanic/Latino	34 (10.8%)	9 (11.7%)	.983
Blood group	-	-	.671
A	123 (34.8%)	34 (41%)	-
B	54 (15.3%)	11 (13.3%)	-
AB	9 (2.5%)	3 (3.6%)	-
O	167 (47.3%)	35 (42.2%)	-
Rhesus factor	-	-	-
+	290 (82.6%)	69 (83.1)	1.0
LTB location	-	-	<.001
Cardiovascular intensive care unit	20 (5.5%)	19 (22.9%)	-
Emergency department	147 (40.2%)	13 (15.7%)	-
Inpatient floor	3 (0.8%)	1 (1.2%)	-
Neonatal intensive care unit	8 (2.2%)	2 (2.4%)	-
Operating room	103 (28.1%)	22 (26.5%)	-
Pediatric intensive care unit	80 (21.9%)	25 (30.1%)	-
Other location	5 (1.4%)	1 (1.2%)	-
Pediatric risk of mortality III (PRISM score)	12.0 [5.0, 22.0]	16.0 [9.0, 22.0]	.08
Resuscitation	-	-	-
Mechanical ventilation	202 (57.9%)	53 (63.9%)	.384
ECMO	20 (5.5%)	16 (19.3%)	<.001
Injury severity score	29.0 [20.0, 38.0]	38.0 [26.0, 41.5]	.427
Vital signs and laboratory values prior to LTB	-	-	-
Age-adjusted hypotension	202 (56.1%)	53 (63.9%)	.245
Tachycardia	207 (61.1%)	56 (67.5%)	.34
Lowest body temperature (°C)	36.1 [35.2, 36.6]	36.2 [35.2, 36.5]	.786
Highest body temperature (°C)	36.8 [36.2, 37.4]	37.2 [36.8, 37.8]	<.001
Lowest pH	7.2 [7.1, 7.3]	7.2 [7.1, 7.3]	.389
Highest serum creatinine (mg/dL)	0.6 [0.4, 0.9]	0.7 [0.4, 1.2]	.03
Lowest hemoglobin (g/dL)	9.8 [7.9, 11.8]	9.0 [7.5, 10.4]	.036
Lowest white blood cell count (K/mm ³)	11.0 [6.3, 18.2]	9.4 [6.0, 16.0]	.304
Lowest platelet count ($\times 10^9/L$)	165.0 [85.5, 258.2]	95.0 [51.0, 203.0]	.003
Highest INR	1.5 [1.2, 2.1]	1.5 [1.2, 1.9]	.853
Highest PT (s)	17.6 [14.9, 23.1]	18.6 [14.6, 21.2]	.872
Highest PTT (s)	42.1 [31.0, 81.1]	51.6 [32.5, 110.6]	.078
Fibrinogen (mg/dL)	184.5 [132.8, 251.2]	215.0 [142.5, 270.0]	.108

TABLE 2 (Continued)

Measures	No AKI (N = 366)	AKI (N = 83)	p-value
Transfused blood products			
Total Volume Transfused (mL/kg)	56.7 [26.7, 108.1]	79.9 [52.0, 193.5]	<.001
pRBCs (mL/kg)	28.6 [14.0, 58.3]	44.0 [19.8, 82.8]	.002
Plasma (mL/kg)	15.3 [4.0, 31.9]	25.1 [9.6, 54.4]	.002
Platelets (mL/kg)	6.0 [0.0, 16.4]	14.3 [4.9, 42.6]	<.001
Received cryoprecipitate (%)	107 (29.2%)	45 (54.2%)	<.001
Plasma:pRBC transfusion ratio	0.5 [0.2, 0.9]	0.6 [0.3, 0.8]	.703
Platelet:pRBC transfusion ratio	0.2 [0.0, 0.4]	0.4 [0.2, 0.6]	<.001

Note: Data presented as either N(%) or median (interquartile range).

Abbreviations: ECMO, extracorporeal membrane oxygenation; INR, international normalized ratio; LTB, life-threatening bleeding; pRBCs, packed red blood cells; PT, prothrombin time; PTT, partial thromboplastin time; Total Volume Transfused, platelets + prbc + plasma + cryoprecipitate.

The results also suggest that the etiology of bleeding is an important factor, and that children with bleeding due to trauma, surgery, and medical reasons may benefit from unique resuscitation strategies. The risk of AKI was higher in children with life-threatening bleeding from operative and medical etiologies relative to children with traumatic bleeding. This may be due to medical comorbidities and underlying illnesses that required surgery or the impact of a surgical procedure versus a traumatic injury.

The possible risk of AKI related to increased platelet to RBC transfusion ratios needs to be considered in the context of the data on traumatic bleeding that demonstrated improved survival. Secondary analyses of randomized trials and meta-analyses of randomized control trials in adults indicate that increased platelet to RBC transfusion ratios are associated with improved survival.^{21,22}

The data regarding associations between hemostatic resuscitation principles with blood components (high ratios or low deficits of plasma and platelets) and outcomes in children has been mixed. Single-center studies without full adjustment for important confounders have not reported independent associations between either high ratios or low deficits with improved survival.^{4,23–25} However, multicenter studies that were able to adjust for important confounders have reported an independent association with improved survival with either higher ratios of plasma to RBCs or low deficits of plasma and platelets.^{1,4,23,26}

Based on the current literature, pediatric general and trauma guidelines suggest the use of high plasma and platelet transfusion ratios for children with life-threatening bleeding.^{27,28} The use of higher ratios in children for traumatic bleeding has become the standard of care, which is reflected in the design of an ongoing trial that is comparing blood components in a 1:1:1 ratio compared to low titer whole blood in children with life-

threatening traumatic bleeding.²⁹ In non-trauma etiologies of life-threatening bleeding, such as medical and operative causes, higher transfusion ratios or lower deficits of plasma and platelets to RBCs have not been associated with improved survival.¹² Therefore, in traumatic life-threatening bleeding, if the use of hemostatic resuscitation principles with blood components does improve survival, then the risk of AKI can be accepted, with the goal of working towards reducing that risk. In non-trauma patients with life-threatening bleeding, if there is no survival benefit and an increased risk of AKI with higher platelet to RBC transfusion ratios then the practice of using platelets liberally in this population needs to be reconsidered.

There are some potential mechanistic hypotheses regarding why increased platelet use in children with life-threatening bleeding increases the risk of AKI. It is possible that the platelets cause a proinflammatory response causing endothelial injury and diffuse microthrombi in the renal vasculature. Cognasse et al. summarized available evidence suggesting that platelet manufacturing can cause platelet injury resulting in morphologic, metabolic, and functional modulations which leads to platelet activation and the release of biological response modifiers (BRMs). As a consequence, platelet concentrates accumulate BRMs during processing and storage, and these BRMs are ultimately transfused alongside platelets and can induce immune responses and posttransfusion reactions in the transfusion recipient.^{30–32} This hypothesis and other mechanisms related to endothelial injury and microthrombi formation need to be explored relative to how platelets are manufactured to include storage temperature.

This study is limited by its retrospective nature and the fact that the database was not designed with this study as the primary objective. There was missing data for some variables and data that was not collected. This

TABLE 3 Demographics of children with life-threatening bleeding with and without acute respiratory distress syndrome (ARDS).

Measures	No ARDS (N = 358)	ARDS (N = 91)	p-value
Sex	-	-	-
Male	199 (55.6%)	48 (52.7%)	.713
Age (years)	7.7 [1.8, 15.0]	5.7 [1.2, 12.8]	.089
Race	-	-	.625
White	190 (63.1%)	52 (65.8%)	-
Black	86 (28.6%)	23 (29.1%)	-
Other	25 (8.3%)	4 (5.1%)	-
Hispanic/Latino	35 (11.3%)	8 (9.8%)	.844
Blood group	-	-	.216
A	121 (35.1%)	36 (39.6%)	-
B	54 (15.7%)	11 (12.1%)	-
AB	7 (2.0%)	5 (5.5%)	-
O	163 (47.2%)	39 (42.9%)	-
Rhesus factor	-	-	-
+	279 (81.3%)	80 (87.9)	.188
LTB location	-	-	.002
Cardiovascular intensive care unit	23 (6.4%)	16 (17.6%)	-
Emergency department	141 (39.4%)	19 (20.9%)	-
Inpatient floor	4 (1.1%)	-	-
Neonatal intensive care unit	9 (2.5%)	1 (1.1%)	-
Operating room	98 (27.4%)	27 (29.7%)	-
Pediatric intensive care unit	78 (21.8%)	27 (29.7%)	-
Other location	5 (1.4%)	1 (1.1%)	-
Pediatric risk of mortality III (PRISM Score)	12.0 [5.0, 22.0]	16.0 [10.0, 22.0]	.019
Resuscitation	-	-	-
Mechanical ventilation	191 (55.4%)	64 (73.6%)	.003
ECMO	21 (5.9%)	15 (16.5%)	.002
Injury severity score	29.0 [20.0, 38.0]	35.0 [26.0, 50.0]	.007
Vital signs and laboratory values prior to LTB	-	-	-
Age-adjusted hypotension	202 (57.4%)	53 (58.2%)	.978
Tachycardia	200 (59.3%)	63 (74.1%)	.017
Lowest body temperature (°C)	36.2 [35.2, 36.6]	36.1 [35.3, 36.4]	.241
Highest body temperature (°C)	36.9 [36.3, 37.5]	37.0 [36.5, 37.7]	.23
Lowest pH	7.2 [7.1, 7.3]	7.2 [7.1, 7.3]	.878
Highest serum creatinine (mg/dL)	0.6 [0.4, 1.0]	0.8 [0.5, 1.2]	.025
Lowest hemoglobin (g/dL)	9.7 [7.8, 11.7]	9.5 [7.6, 11.9]	.871
Lowest white blood cell count (K/mm ³)	10.4 [5.9, 16.8]	10.9 [7.4, 19.6]	.229
Lowest platelet count ($\times 10^9/L$)	153.0 [76.0, 244.0]	151.5 [82.5, 255.2]	.645
Highest INR	1.5 [1.2, 2.2]	1.5 [1.3, 1.8]	.595
Highest PT (s)	17.8 [14.9, 24.6]	17.3 [14.9, 20.0]	.347
Highest PTT (s)	44.0 [32.0, 91.0]	35.7 [30.1, 75.3]	.19
Fibrinogen (mg/dL)	186.0 [131.8, 255.2]	201.0 [146.5, 269.0]	.159

TABLE 3 (Continued)

Measures	No ARDS (N = 358)	ARDS (N = 91)	p-value
Transfused blood products			
Total Volume Transfused (mL/kg)	56.9 [28.1, 114.1]	68.2 [37.4, 151.6]	.038
pRBCs (mL/kg)	29.1 [15.4, 61.1]	34.9 [18.3, 68.6]	.13
Plasma (mL/kg)	15.7 [4.0, 34.2]	21.2 [9.0, 42.1]	.051
Platelets (mL/kg)	6.8 [0.0, 17.6]	11.1 [0.0, 26.4]	.011
Received cryoprecipitate (%)	113 (31.6%)	39 (42.9%)	.056
Plasma:pRBC transfusion ratio	0.5 [0.2, 0.9]	0.6 [0.3, 0.9]	.339
Platelet:pRBC transfusion ratio	0.2 [0.0, 0.4]	0.3 [0.1, 0.5]	.003

Note: Data presented as either N(%) or median (interquartile range).

Abbreviations: ECMO, extracorporeal membrane oxygenation; INR, international normalized ratio; LTB, life-threatening bleeding; pRBCs, packed red blood cells; PT, prothrombin time; PTT, partial thromboplastin time; Total Volume Transfused, platelets + prbc + plasma + cryoprecipitate.

TABLE 4 Weighted logistic regression model for acute kidney injury (AKI).

Characteristic	Odds ratio	95% confidence interval	p-value
Platelet deficit	0.99	0.99, 1.00	.065
Plasma:RBC ratio (per 0.1 units increase)	0.44	0.17, 1.14	.089
Platelet:RBC ratio (per 0.1 units increase)	1.127	1.025, 1.239	.013
Mechanical ventilation prior to LTB event			
No	Reference	-	
Yes	1.01	0.41, 2.52	.984
Clinical group			
Trauma patients	Reference	-	
Medical patients	2.80	1.27, 6.17	.011
Operative patients	3.32	1.27, 8.74	.015
Sex			
Female	Reference	-	
Male	1.37	0.7, 2.67	.359
ECMO			
No	Reference	-	
Yes	1.03	0.36, 2.94	.952
Platelet count	1.00	0.99, 1.00	.142
Cardiac arrest			
No	Reference	-	
Yes	2.19	0.61, 7.84	.227
PRISM score	0.95	0.91, 1.01	.081
Age (years)	1.01	0.95, 1.07	.754
Total Volume Transfused (per 10 mL/kg)	1.04	1.01, 1.07	.006

Note: The number of subjects included in the model is 351 after excluding subjects with missing values. Model diagnostics: passed Hosmer and Lemeshow goodness of fit test, p-value = .111.

Abbreviations: ECMO, extracorporeal membrane oxygenation; LTB, life-threatening bleeding; pRBCs, packed red blood cells; PRISM, pediatric risk of mortality score; RBC, red blood cell; Total Volume Transfused, platelets + pRBC + plasma + cryoprecipitate.

may have led to not including all relevant variables in the adjusted models for AKI and ARDS. It is possible that subjects' underlying conditions increased the risk of AKI and could not be adjusted for in our adjusted model. The

small sample size per bleeding etiology did not allow for performing adjusted analyses in each specific group.

In summary, the use of increased platelet to RBCs transfusion ratios in children with life-threatening

TABLE 5 Weighted logistic regression model for acute respiratory distress syndrome (ARDS).

Characteristic	Odds ratio	95% confidence interval	p-value
Platelet deficit	1.00	1.00, 1.01	.179
Plasma:RBC ratio (per 0.1 units increase)	1.014	0.993, 1.035	.196
Platelet:RBC ratio (per 0.1 units increase)	1.053	0.977, 1.135	.176
Mechanical ventilation prior to LTB event			
No	Reference	-	
Yes	1.89	0.85, 4.24	.120
Clinical group			
Trauma patients	Reference	-	
Medical patients	1.00	0.46, 2.18	.998
Operative patients	0.77	0.33, 1.80	.542
Sex			
Female	Reference	-	
Male	1.23	0.63, 2.40	.547
ECMO			
No	Reference	-	
Yes	2.06	0.77, 5.51	.148
Platelet count	-	-	-
Cardiac arrest	-	-	-
PRISM score	0.99	0.95, 1.02	.435
Age (years)	0.98	0.94, 1.03	.468

Note: The number of subjects included in the model is 351 after excluding subjects with missing values. Model diagnostics: passed Hosmer and Lemeshow goodness of fit test, p -value = .6561.

Abbreviations: ECMO, extracorporeal membrane oxygenation; LTB, life-threatening bleeding; PRISM, pediatric risk of mortality score; RBC, red blood cell.

bleeding of any etiology may increase the risk of AKI while not increasing the risk of ARDS. The empiric use of platelets may not be beneficial for all etiologies of bleeding. The development of accurate point-of-care tests that rapidly determine the indication for platelet transfusion may allow for precision transfusion medicine where the benefits of platelet transfusion can be achieved while minimizing risk. Trials examining the efficacy and safety of platelet transfusions that incorporate biologically driven inclusion criteria and indications for platelets are needed in children with life-threatening bleeding.

CONFLICT OF INTEREST STATEMENT

No COI disclosures for any authors except CDJ who discloses financial support as a consultant at Westat, Medtronic, and Immucor and PCS who is a consultant for Cerus and Hemanext, on the advisory board for Octapharma and Haima, and is a Co-Founder and CMO for Kalocyte.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Vaizer R, Leeper CM, Lu L, Josephson CD, Leonard JC, Brown JB, et al. Increased platelet to red blood cell transfusion ratio associated with acute kidney injury in children with life-threatening bleeding. *Transfusion.* 2024;64(S2):S62–71. <https://doi.org/10.1111/trf.17788>